CENTOS DHCP AND DNS SERVER ADMINISTRATION LAB

A Practical Linux Network Services Project

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Program

Bachelor of Science in Information Technology (BSIT)

Project Category

Linux Server Administration & Network Services

Platform:	**VMware Workstation**
Operating System:	**CentOS Linux**
Network Services:	**DHCP Server, DNS Server**
Domain:	**lab.local**
Network:	**192.168.10.0/24**

PROJECT STATUS

✔️ **COMPLETED**

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1. Introduction

This project documents the design, configuration, and testing of a Dynamic Host Configuration Protocol (DHCP) server and a Domain Name System (DNS) server on CentOS Linux. The lab was carried out in a virtualized environment using VMware Workstation to simulate a small enterprise network segment. The purpose of the exercise is to demonstrate practical, hands-on competency in configuring and administering the two network services that most commonly underpin IP address management and name resolution in a Linux-based infrastructure.

The DHCP service was configured to automatically assign IP addresses, subnet information, default gateway, and DNS server details to client machines on the 192.168.10.0/24 network. The DNS service was configured using BIND (Berkeley Internet Name Domain) to provide both forward and reverse lookup resolution for the lab.local domain.

2. Objectives

The primary objectives of this lab were to:

- Install and configure the ISC DHCP server on CentOS Linux.
- Design an IP addressing scheme for the 192.168.10.0/24 network.
- Configure DHCP scopes, lease times, and reserved options (gateway, DNS, domain name).
- Install and configure the BIND DNS server to support the lab.local domain.
- Create and validate forward and reverse lookup zone files.
- Configure the CentOS firewall (firewalld) to permit DHCP and DNS traffic.
- Test and verify DHCP address assignment and DNS name resolution from client machines.
- Document common troubleshooting steps for both services.

3. Lab Environment and Prerequisites

The lab was built using the following virtualized environment and software components:

Component	Specification
Virtualization Platform	VMware Workstation
Server Operating System	CentOS Linux (Minimal Install)
Client Operating System	CentOS / Windows Client (for testing)
Network Adapter Mode	Host-Only / Custom VMnet
Server Hostname	dns-dhcp-server.lab.local

Component	Specification
Domain Name	lab.local
Network Address	192.168.10.0/24
Server Static IP	192.168.10.1
DHCP Scope Range	192.168.10.100 – 192.168.10.200
Default Gateway	192.168.10.1
DNS Server Address	192.168.10.1

Prerequisites

- A working CentOS Linux installation with root or sudo access.
- A static IP address configured on the server's network interface.
- Internet or local repository access for package installation via yum/dnf.
- Basic familiarity with the Linux command line and a text editor (vi/nano).
- SELinux and firewalld enabled (default CentOS security posture).

4. Network Topology

The lab network consists of a single CentOS server hosting both the DHCP and DNS roles, connected to a private virtual network segment. Client machines obtain their network configuration automatically from the DHCP server, which in turn directs them to the DNS server for name resolution within the lab.local domain.

Device	Role	IP Address	Notes
dns-dhcp-server.lab.local	DHCP + DNS Server	192.168.10.1 (Static)	Single combined services host
client-01.lab.local	DHCP Client	192.168.10.101 (Dynamic)	Leased from DHCP pool
client-02.lab.local	DHCP Client	192.168.10.102 (Dynamic)	Leased from DHCP pool

> **Note:** All virtual machines were attached to the same isolated VMnet to prevent interference with production or home network DHCP/DNS services.

5. DHCP Server Configuration

5.1 Installing the DHCP Server Package

The ISC DHCP server package was installed using the following command:

```
yum install dhcp-server -y
```

5.2 Configuring a Static IP Address

Before configuring DHCP, the server's own network interface was assigned a static IP address by editing the interface configuration file:

```
vi /etc/sysconfig/network-scripts/ifcfg-ens160
```

```
BOOTPROTO=static
IPADDR=192.168.10.1
PREFIX=24
GATEWAY=192.168.10.1
DNS1=192.168.10.1
ONBOOT=yes
```

The network service was then restarted to apply the changes:

```
systemctl restart NetworkManager
```

5.3 Editing dhcpd.conf

The main DHCP configuration file, /etc/dhcp/dhcpd.conf, was edited to define the subnet, address pool, lease times, and network options:

```
option domain-name "lab.local";
option domain-name-servers 192.168.10.1;
default-lease-time 600;
max-lease-time 7200;
authoritative;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option domain-name-servers 192.168.10.1;
    option domain-name "lab.local";
}
```

5.4 Starting and Enabling the DHCP Service

```
systemctl start dhcpd
```

```
systemctl enable dhcpd
systemctl status dhcpd
```

5.5 Configuring the Firewall for DHCP

Since firewalld is enabled by default on CentOS, the DHCP service was permitted through the firewall:

```
firewall-cmd --permanent --add-service=dhcp
firewall-cmd --reload
```

6. DNS Server Configuration

6.1 Installing BIND

The BIND DNS server and its utilities were installed using:

```
yum install bind bind-utils -y
```

6.2 Configuring named.conf

The main BIND configuration file, /etc/named.conf, was updated to listen on the server's interface and to define the lab.local zone:

```
listen-on port 53 { 127.0.0.1; 192.168.10.1; };
allow-query     { localhost; 192.168.10.0/24; };
recursion yes;

zone "lab.local" IN {
    type master;
    file "lab.local.zone";
    allow-update { none; };
};

zone "10.168.192.in-addr.arpa" IN {
    type master;
    file "10.168.192.rev";
    allow-update { none; };
};
```

6.3 Creating the Forward Lookup Zone File

The forward zone file was created at /var/named/lab.local.zone to resolve hostnames to IP addresses:

```
$TTL 86400
@   IN  SOA   dns-dhcp-server.lab.local. admin.lab.local. (
        2026072901 ; Serial
```

```
        3600       ; Refresh
        1800       ; Retry
        604800     ; Expire
        86400 )    ; Minimum TTL

@               IN   NS    dns-dhcp-server.lab.local.
dns-dhcp-server IN   A     192.168.10.1
client-01       IN   A    192.168.10.101
client-02       IN   A    192.168.10.102
```

6.4 Creating the Reverse Lookup Zone File

The reverse zone file was created at /var/named/10.168.192.rev to resolve IP addresses back to hostnames:

```
$TTL 86400
@   IN  SOA   dns-dhcp-server.lab.local. admin.lab.local. (
        2026072901 ; Serial
        3600       ; Refresh
        1800       ; Retry
        604800     ; Expire
        86400 )    ; Minimum TTL

@   IN  NS    dns-dhcp-server.lab.local.
1   IN  PTR   dns-dhcp-server.lab.local.
101 IN  PTR   client-01.lab.local.
102 IN  PTR   client-02.lab.local.
```

6.5 Setting Permissions and Validating Zone Syntax

File ownership was set to the named user and group, and the zone files were validated for syntax errors before starting the service:

```
chown named:named /var/named/lab.local.zone /var/named/10.168.192.rev
chmod 640 /var/named/lab.local.zone /var/named/10.168.192.rev
named-checkconf /etc/named.conf
named-checkzone lab.local /var/named/lab.local.zone
named-checkzone 10.168.192.in-addr.arpa /var/named/10.168.192.rev
```

6.6 Starting and Enabling the DNS Service

```
systemctl start named
systemctl enable named
systemctl status named
```

6.7 Configuring the Firewall for DNS

```
firewall-cmd --permanent --add-service=dns
```

```
firewall-cmd --reload
```

7. Testing and Verification

7.1 Verifying DHCP Lease Assignment

On a client machine, the network interface was released and renewed to request a new lease from the DHCP server:

```
dhclient -r
dhclient
ip addr show
```

The client successfully received an IP address within the configured pool (192.168.10.100–192.168.10.200), along with the correct subnet mask, gateway, and DNS server.

7.2 Verifying Active DHCP Leases on the Server

Active leases were confirmed on the server by inspecting the lease database:

```
cat /var/lib/dhcpd/dhcpd.leases
```

7.3 Verifying Forward DNS Resolution

```
dig client-01.lab.local
nslookup client-01.lab.local 192.168.10.1
```

Both commands returned the correct A record, confirming that the forward lookup zone was functioning correctly.

7.4 Verifying Reverse DNS Resolution

```
dig -x 192.168.10.101
```

The reverse lookup query returned the corresponding PTR record (client-01.lab.local), confirming correct configuration of the reverse zone.

7.5 Summary of Test Results

Test	Expected Result	Result
Client receives DHCP lease	IP in range 192.168.10.100–200	Passed
Correct gateway/DNS assigned	Gateway and DNS = 192.168.10.1	Passed
Forward lookup (A record)	Resolves hostname to IP	Passed

Test	Expected Result	Result
Reverse lookup (PTR record)	Resolves IP to hostname	Passed
Services start on boot	dhcpd and named enabled	Passed

8. Troubleshooting Guide

The following issues are commonly encountered during DHCP and DNS configuration on CentOS, along with the steps used to resolve them in this lab:

Issue	Likely Cause	Resolution
Client does not receive an IP address	dhcpd service not running or firewall blocking traffic	Check systemctl status dhcpd and confirm firewall-cmd --add-service=dhcp was applied
dhcpd fails to start	Syntax error in dhcpd.conf	Run dhcpd -t to test configuration syntax
named fails to start	Zone file syntax error or incorrect permissions	Run named-checkzone and verify ownership is named:named
nslookup times out	Firewall blocking port 53 or wrong DNS server set	Confirm firewall-cmd --add-service=dns and correct resolv.conf entry
Reverse lookup fails	Incorrect PTR record or zone name mismatch in named.conf	Verify the in-addr.arpa zone name matches the network in reverse notation
SELinux denies named access to zone files	SELinux context on zone files incorrect	Run restorecon -Rv /var/named to reset the correct context

9. Conclusion

This lab successfully demonstrated the end-to-end configuration of a DHCP and DNS server on CentOS Linux within a VMware Workstation environment. The DHCP server was configured to automatically assign IP addresses, subnet details, and DNS information to client machines on the 192.168.10.0/24 network, while the BIND-based DNS server provided reliable forward and reverse name resolution for the lab.local domain.

Through this project, practical experience was gained in Linux service configuration, firewall management, zone file creation, and network troubleshooting — core competencies required for Linux server and network administration roles. The skills demonstrated here form a foundation for more advanced topics such as DNS security (DNSSEC), DHCP failover, and integration with Active Directory or FreeIPA environments.

10. Appendix: Command Reference

A quick reference of the key commands used throughout this lab:

Command	Purpose
yum install dhcp-server -y	Install the DHCP server package
yum install bind bind-utils -y	Install the BIND DNS server and utilities
systemctl start/enable dhcpd	Start and enable the DHCP service
systemctl start/enable named	Start and enable the DNS service
firewall-cmd --add-service=dhcp/dns	Allow DHCP/DNS traffic through firewalld
named-checkconf / named-checkzone	Validate BIND configuration and zone files
dhclient -r / dhclient	Release and renew a DHCP lease on a client
dig / nslookup	Query DNS records for forward and reverse lookups
cat /var/lib/dhcpd/dhcpd.leases	View active DHCP lease assignments